

Below code snippet and glimpse of Taxi Data Analysis And Dashboard In Python

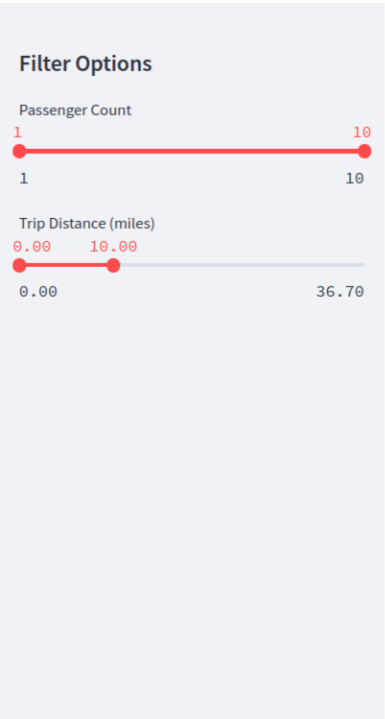
```
# Trip Duration Distribution
st.subheader('Trip Duration Distribution')
fig, ax = plt.subplots(figsize=(10, 6))
sns.histplot(filtered_df['trip_duration'], kde=True, color='blue', bins=30, ax=ax)
st.pyplot(fig)

# Passengers by Hour
st.subheader('Number of Passengers by Hour')
fig, ax = plt.subplots(figsize=(10, 6))
sns.countplot(data=filtered_df, x='hour', palette='viridis', ax=ax)
st.pyplot(fig)

# Fare vs Distance
st.subheader('Fare vs Distance')
fig, ax = plt.subplots(figsize=(10, 6))
sns.scatterplot(data=filtered_df, x='distance', y='fare', hue='payment', palette='coolwarm', ax=ax)
st.pyplot(fig)

# Fare by Month
st.subheader('Fare Distribution by Month')
fig, ax = plt.subplots(figsize=(10, 6))
sns.boxplot(data=filtered_df, x='month', y='fare', palette='muted', ax=ax)
st.pyplot(fig)

# Correlation Heatmap (fix applied here)
st.subheader('Correlation Heatmap')
fig, ax = plt.subplots(figsize=(12, 8))
numeric_df = filtered_df.select_dtypes(include='number') # Ensure only numeric columns
sns.heatmap(numeric_df.corr(), annot=True, cmap='coolwarm', fmt='.2f', linewidths=0.5, ax=ax)
st.pyplot(fig)
```



Taxi Trip Analysis Dashboard

Filtered Data Shape: (5903, 18)

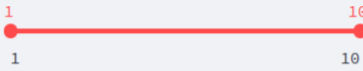
Sample of Filtered Data:

	pickup	dropoff	passengers	distance	fare	tip	tolls	total	color
0	2019-03-23 20:21:00	2019-03-23 20:27:00	1	1.6	7	2.15	0	12.95	yellow
1	2019-03-04 16:11:00	2019-03-04 16:19:00	1	0.79	5	0	0	9.3	yellow
2	2019-03-27 17:53:00	2019-03-27 18:00:00	1	1.37	7.5	2.36	0	14.16	yellow
3	2019-03-10 01:23:00	2019-03-10 01:49:00	1	7.7	27	6.15	0	36.95	yellow
4	2019-03-30 13:27:00	2019-03-30 13:37:00	3	2.16	9	1.1	0	13.4	yellow

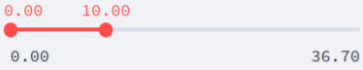
	pickup	dropoff	passengers	distance	fare	tip	tolls	total	color
6427	2019-03-23 18:26:00	2019-03-23 18:49:00	1	7.07	20	0	0	20	green
6428	2019-03-31 09:51:00	2019-03-31 09:55:00	1	0.75	4.5	1.06	0	6.36	green
6430	2019-03-23 22:55:00	2019-03-23 23:14:00	1	4.14	16	0	0	17.3	green
6431	2019-03-04 10:09:00	2019-03-04 10:14:00	1	1.12	6	0	0	6.8	green

Filter Options

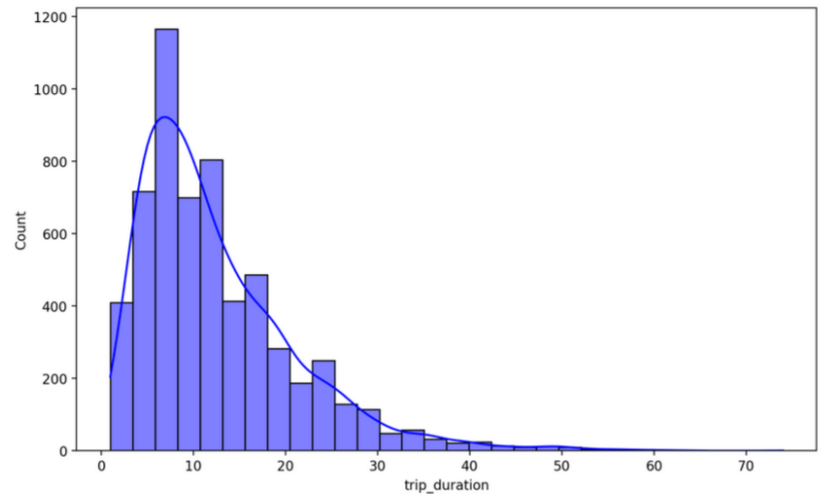
Passenger Count



Trip Distance (miles)



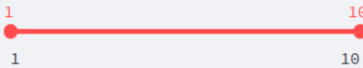
Trip Duration Distribution



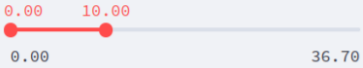
trip_duration

Filter Options

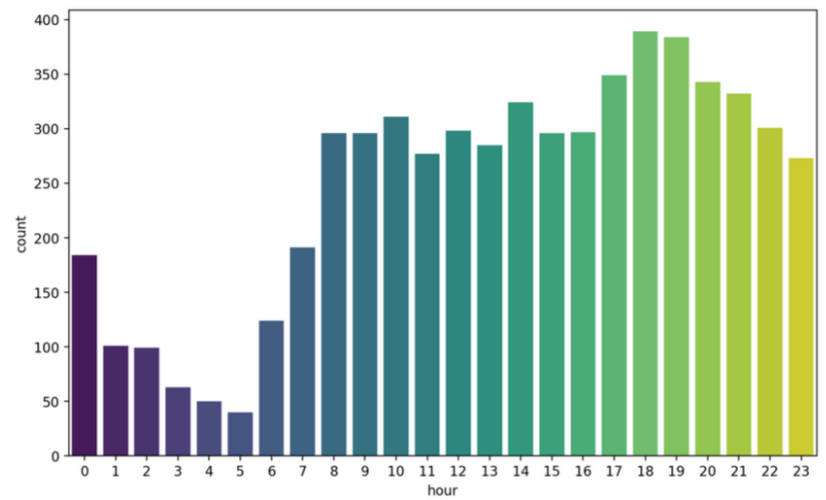
Passenger Count



Trip Distance (miles)



Number of Passengers by Hour



Filter Options

Passenger Count

1

10

1

10

Trip Distance (miles)

0.00

10.00

0.00

36.70

Filter Options

Passenger Count

1

10

1

10

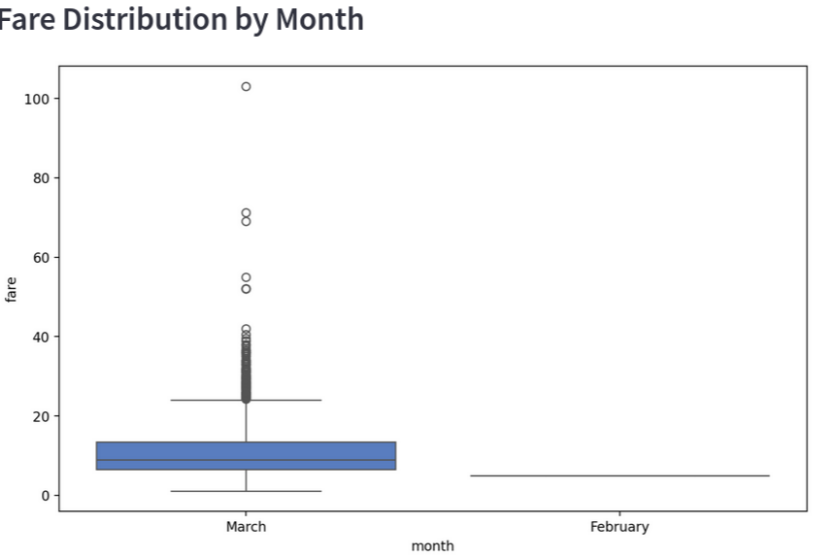
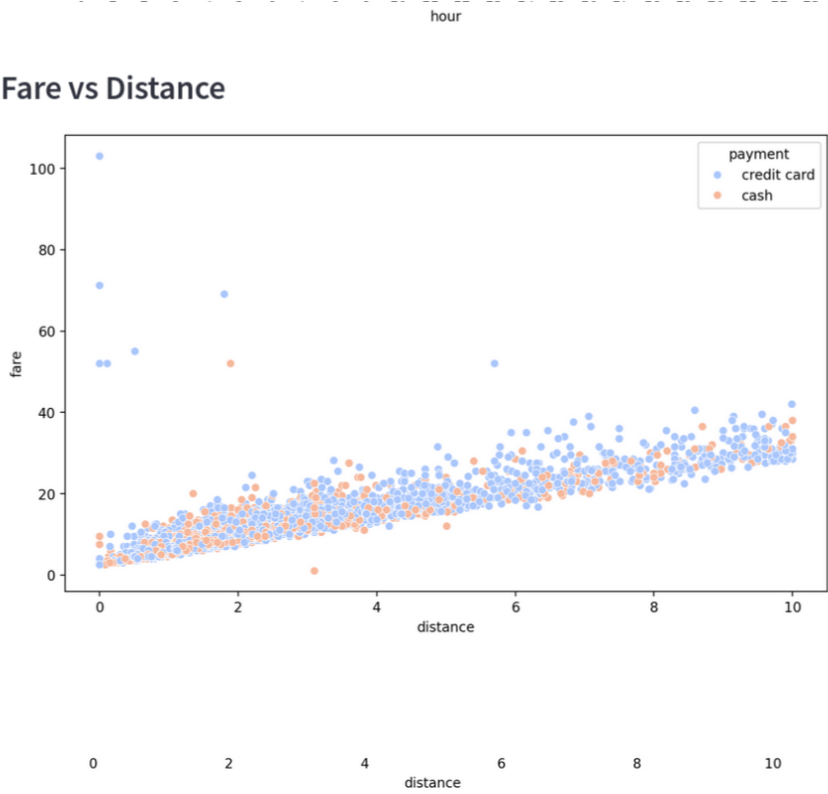
Trip Distance (miles)

0.00

10.00

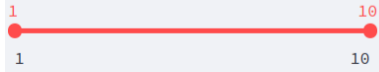
0.00

36.70

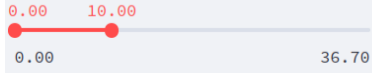


Filter Options

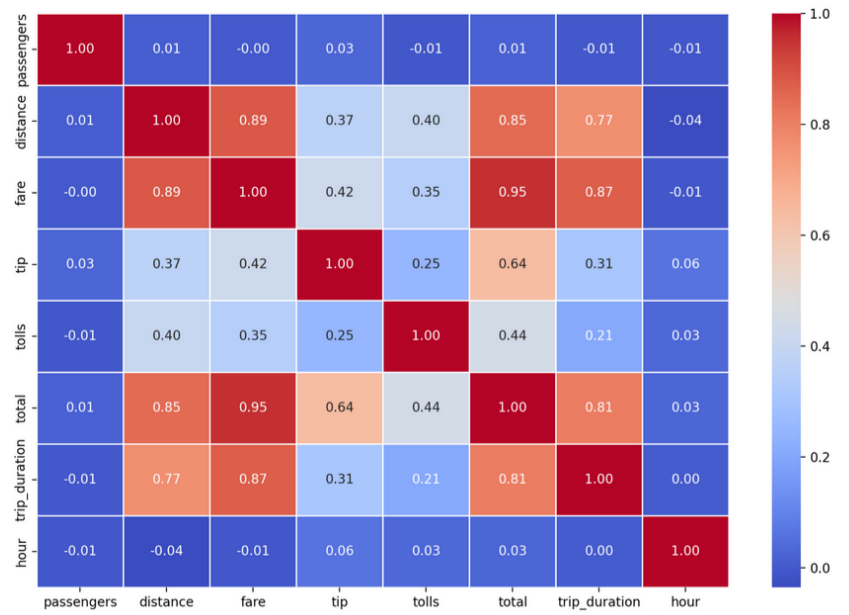
Passenger Count



Trip Distance (miles)



Correlation Heatmap



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